Procurement Restructuring & Supplier Intelligence Dossier

Tender UP-62A2197E - Construction of viaduct and elevated station building (Pkg-2) (for 13.3 kms out of total 28.5 kms)
Generated 2026-11-13T04:41:32+00:00 by TenderShield Nexus (decision support - not an award decision)

Keep the procurement as one integrated contract: splitting does not improve procurement value under the stated weights.

A. Original procurement

Estimated value: unknown. Current structure: 1 package(s): Civil & building construction.

ID	Component	Items	Value	Value basis	Source pages
C1	Civil & building construction	1	unknown	unknown	1

B. Proposed restructuring

1 package(s): Civil & building construction

C. SME opportunity map

Pkg	Scope	Value	Turnover req.	Capable firms	Meet scale rule	SME candidates	Bidders (ind.)
P1	Civil & building construction	unknown	unknown	1119	0	795	2.5

P1 potential supplier pool (top 6 of 1119)

Supplier	Country	Status	MSME status	Score	Evidence
A. K. PATEL AND COMPANY [CONSTRUCTION] PVT. LTD., SURAT	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.775	6 awards in Civil & building construction (largest [18.449])
AMAR CONSTRUCTION CO.	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_LARGE	0.775	5 awards in Civil & building construction (largest [33.149])
BACKBONE ENTERPRISES LTD.	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_LARGE	0.775	7 awards in Civil & building construction (largest [371.604])
BHOLA SINGH JAI PRAKASH & CO	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.775	5 awards in Civil & building construction (largest [4.644])
M/S CEEKAY ASSOC. PVT. LTD.	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.775	6 awards in Civil & building construction (largest [5.646])
M/S CHANDRAGIRI CONSTRUCTION	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_LARGE	0.775	8 awards in Civil & building construction (largest [37.128])

D. Supplier relationships (secondary objective)

10 shortlisted suppliers screened; 0 relationship(s) require review. Signals for human review only. Relationships are not evidence of wrongdoing.

Supplier A	Supplier B	Observed link	Review score	Ordinary explanations
A. K. PATEL AND COMPANY [CONSTRUCTION] PVT. LTD., SURAT	AMAR CONSTRUCTION CO.	same_project	0.1	Large projects legitimately award separate packages to different firms
AMAR CONSTRUCTION CO.	BACKBONE ENTERPRISES LTD.	same_project	0.1	Large projects legitimately award separate packages to different firms
AMAR CONSTRUCTION CO.	M/S CHANDRAGIRI CONSTRUCTION	same_project	0.1	Large projects legitimately award separate packages to different firms
BACKBONE ENTERPRISES LTD.	M/S CHANDRAGIRI CONSTRUCTION	same_project	0.1	Large projects legitimately award separate packages to different firms
BHOLA SINGH JAI PRAKASH & CO	M/S CEEKAY ASSOC. PVT. LTD.	same_project	0.1	Large projects legitimately award separate packages to different firms

Supplier A	Supplier B	Observed link	Review score	Ordinary explanations
BHOLA SINGH JAI PRAKASH & CO	M/S GANGOTRI ENTERPRISES LTD., LKO	same_project	0.1	Large projects legitimately award separate packages to different firms
BHOLA SINGH JAI PRAKASH & CO	M/S HIRA SINGH SOBTI CONTRACTS LTD.	same_project	0.1	Large projects legitimately award separate packages to different firms
M/S CEEKAY ASSOC. PVT. LTD.	M/S GANGOTRI ENTERPRISES LTD., LKO	same_project	0.1	Large projects legitimately award separate packages to different firms
M/S CEEKAY ASSOC. PVT. LTD.	M/S HIRA SINGH SOBTI CONTRACTS LTD.	same_project	0.1	Large projects legitimately award separate packages to different firms
M/S GANGOTRI ENTERPRISES LTD., LKO	M/S HIRA SINGH SOBTI CONTRACTS LTD.	same_project	0.1	Large projects legitimately award separate packages to different firms
FCI ARAVALI GYPSUM MINERALS INDIA LTD.	M/S GANGOTRI ENTERPRISES LTD., LKO	same_project	0.1	Large projects legitimately award separate packages to different firms

E. Scenario comparison

Scenario	Packages	Score	Competition	Coverage	SME access	Coordination	Integration risk	Scale loss
A	1	0.35	0.5	1.0	0	0.0	0.0	0
B	1	0.35	0.5	1.0	0	0.0	0.0	0
C	1	0.35	0.5	1.0	0	0.0	0.0	0

Weight sensitivity: the recommended structure stays best in 100% of 300 runs with every weight perturbed (each weight x U(0.5, 1.5)). Weights: {"competition": 0.2, "coordination": 0.1, "coverage": 0.25, "integration_risk": 0.15, "scale_loss": 0.05, "sme_access": 0.25}

F. Recommendation

Decision support only. The procurement authority decides the structure, verifies supplier evidence and MSME status, and reviews any flagged relationships.

[WARNING] Component values unknown - Upload the BOQ, or enter the tender's estimated value

[INFO] MSME status shown as a size proxy unless Udyam evidence was imported - Import verified Udyam records via POST /api/market/udyam

Data sources and models

Market scope: ['India']; 7434 award records in scope from worldbank_awards:major_contract_awards:india, prozorro:ua.

capability_classifier: TF-IDF (word 1-2g + char 3-5g) + multinomial logistic regression, lexicon-blended at inference. Trained 2026-11-13T04:35:49+00:00 on Common Procurement Vocabulary (CPV 2008) (9454 rows); Annotated tender lines (53 rows). Metrics: {"cv_accuracy": 0.7875, "cv_folds": 5, "cv_macro_f1": 0.7383}

- CPV descriptions are short catalogue phrases; long Indian BOQ lines are a domain shift - the lexicon blend and the annotated-line evaluation exist for this reason.

- Single label per text; a mixed BOQ line is assigned its dominant capability.

competition_model: HistGradientBoostingRegressor (Poisson loss) on log10(value Rs Cr) + capability one-hot. Trained 2026-11-13T04:35:51+00:00 on Prozorro lots (Ukraine public procurement, 2019-2023) (92438 rows). Metrics: {"mae": 1.0558, "mae_baseline": 1.064, "mean_bidders_test": 1.979, "poisson_deviance": 0.6907, "poisson_deviance_baseline": 0.7008, "split": "time: train 2019-2021, test 2022+", "test_rows": 14170, "train_rows": 78268}

- Learned on Ukrainian open tenders; used as a relative indicator between scenarios, not a forecast of Indian bidder counts. Recalibrate with CPPP bid-opening data.

- Lots in the data carry one CPV division, so the model has no notion of bundling; the bundling penalty comes from real supplier co-capability evidence instead.

- Observational data: larger lots in this data attract slightly more bidders; that is a correlation (buyers, procedures and markets differ), not a causal effect of lot size.

Capability taxonomy keys: civil_works = Civil & building construction, electrical = Electrical systems & power, signalling_control = Signalling, traffic & safety control, instrumentation_sensors = Instrumentation, sensors & metering, ict_software = Software, ICT & control systems, telecom_networks = Telecom & communication networks, mechanical_hvac_plumbing = Mechanical, HVAC, pumps & plumbing, industrial_machinery = Industrial & process machinery, structural_fabrication = Structural steel & specialised fabrication, consultancy_design = Design, engineering & supervision services, operation_maintenance = Operation & maintenance services, vehicles_equipment = Vehicles & heavy equipment, medical_lab = Medical & laboratory equipment, other_goods_services = Other goods & services